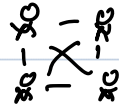


Computer Networks

↓
electronic
device

group of interconnected entities

social network



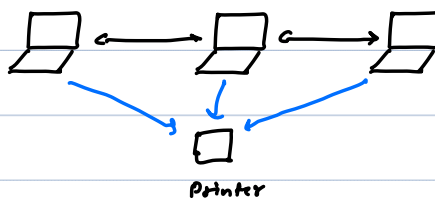
transportation network

roads, railways, airways

computer networks

computer and devices

- it is network of interconnected computers.



- to communicate to each other.
- to exchange data
- to share resources

Internet

U.S. department of defense.

ARPA → advanced research project Agency

↳ ARPANET (now it is internet)

0 0
0 0

0 0
0 0



AlgoPrep

us

0

0

china

India

0

0

germany

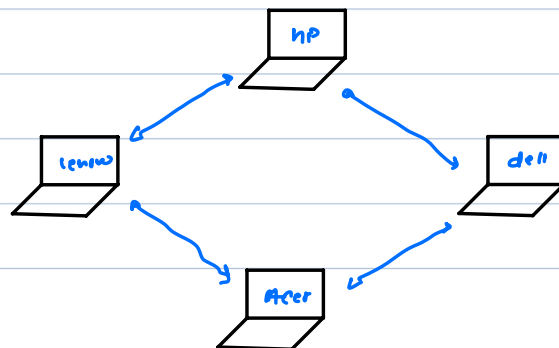
language barrier

english

↓

rules

to communicate to each other we need
something common between them.



to communicate to each other we have to follow rules / protocols.

http

tcp

ip

UDP

FTP

IE TF - (Internet engineering task force)

RFCs (Request for comments)

It will be published on RFCs

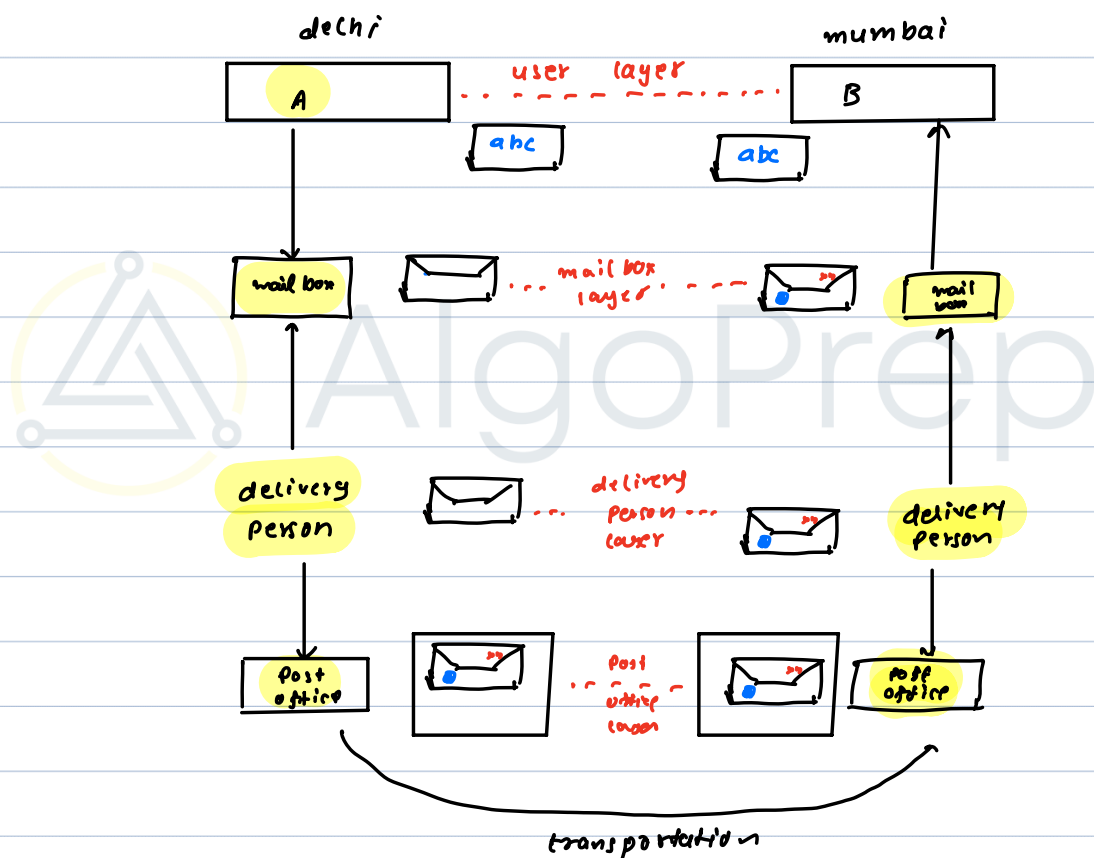


AlgoPrep

layering architecture

it is used to design protocols.

postal network



- each layer is responsible for their own work.
- we divide the task into smaller and manageable layers.
- its provide security.
- each layer do their work and forward to next layer

OSI model (open system interconnection)

How the data is flow

application
layer

creation of data
and send to the another
user.

presentation
layer

we format it, encrypt, compress
this is understood by receiver.

session
layer

we are established the
session.

transport
layer

it ensure end to end
communication

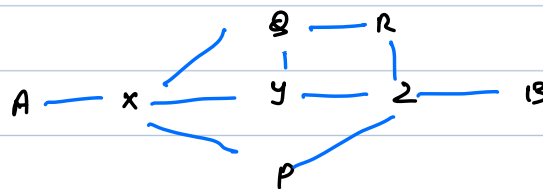
$(\overset{1}{abc} | \overset{2}{xyz} | \overset{3}{wxy})$ x byte

→ serialize

→ tracking

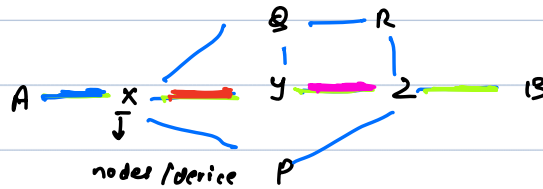
network
layer

its decide which route
is best.



there are many ways

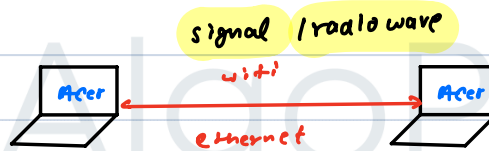
data link



it manages nodes to nodes communication.

Physical
layer

send the data into signal /
radio waves



break till 9:30

address

pt.no-102, st-1, landmark,
city name, name, pt.no.

↓ structured way

House no, street, area, landmark, city

IP address

ipv4

ipv6

32 bits

$\frac{8 \text{ bit}}{(0-255)} \cdot \frac{8 \text{ bit}}{(0-255)} \cdot \frac{8 \text{ bit}}{(0-255)} \cdot \frac{8 \text{ bit}}{(0-255)}$
 $198 \cdot 168 \cdot 1 \cdot 10$
 $198 \cdot 170 \cdot 150 \cdot 259 \quad \times$

total combination 2^{32} 4.3 billion

currently we have more than 4.3 b devices

ipv6

128 bits

↳ its divide into eight part.

$\frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot} \cdot \frac{16 \text{ bit}}{\cdot}$

total combination 2^{128} → 340 trillion

it is very huge number

still ipv4 is more popular

101	102
201	102

1

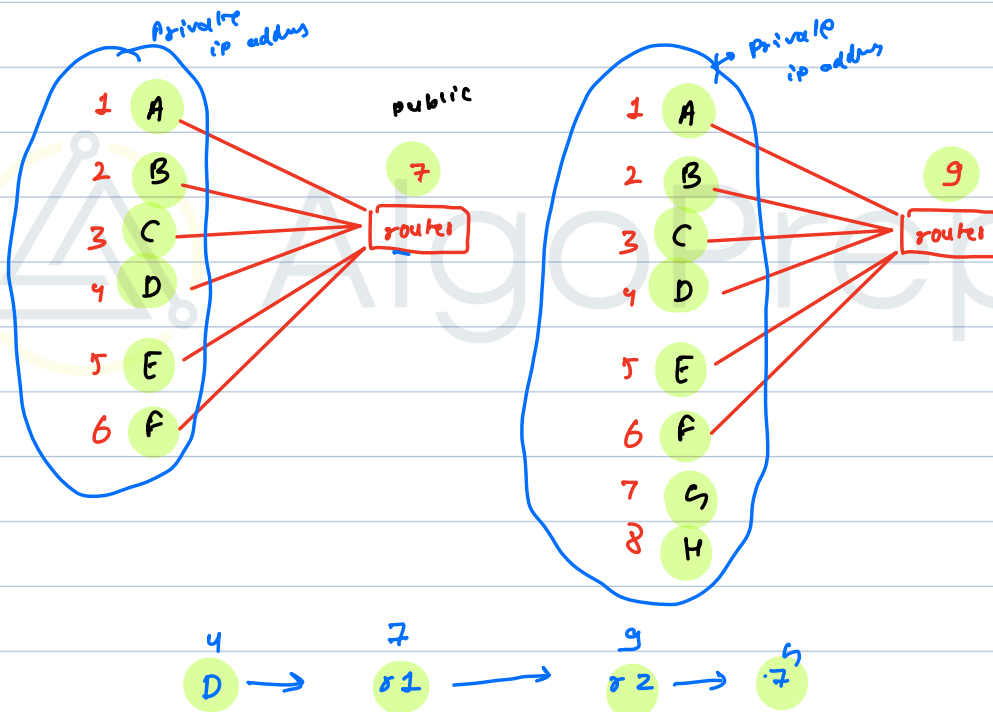
101	102
201	102
3	1

2

101	102
201	102
1	2

3

key → hotel no → room no.



total device = 16

ip address = 9

NAT (network address translation)

it mapping private to public ip address

Internet service provider (ISP)